Formal Languages and Finite Automata Homework #3

1.

a.

$$S$$

$S \to aSAB \mid aa$

$$\Rightarrow aSAB$$

$S \to aSAB \mid aa$

$$\Rightarrow aaaAB$$

$A \to B$

$$\Rightarrow aaaBB$$

$B \to bbB \mid bb$

$$\Rightarrow aaabbB$$

$B \to bbB \mid bb$

$$\Rightarrow aaabbbbB$$

$B \to bbB \mid bb$

$$\Rightarrow aaabbbbbb$$

b.

$$S$$

$S \to aSAB \mid aa$

$$\Rightarrow aSAB$$

$B \to bbB \mid bb$

$$\Rightarrow aSABbbB$$

$B \to bbB \mid bb$

$$\Rightarrow aSAbbbb$$

$A \to B$

$$\Rightarrow aSBbbbb$$

$B \to bbB \mid bb$

$$\Rightarrow aSbbbbbb$$

$S \to aSAB \mid aa$

$$\Rightarrow aaabbbbbb$$

c.

```
          S
       /  /  \  \
     a  S     A    B
       /  \    |    |  \  \
     a      a  B    b    b    B
              /  \           /  \
             b      b       b      b
```

2.

a.

$$S \to AB$$

$$A \to aA \mid \lambda$$

$$B \to bB \mid \lambda$$

b.

$$S \to aSc \mid A$$

$$A \to aAb \mid \lambda$$

3.

Let

$$w = ibtibtaea$$

We can get two different trees.

a.

```
      S
   / / \ \
  i E   t  S
    |     / | \ \ \
    b   i  E t S e S
           |   |   |
           b   a   a
```

b.

```
          S
     / / / \ \ \
    i E t S e   S
      |  /|\    |
      b i E t S a
          |   |
          b   a
```

They get the same result, but with different meanings.

4.

$$S \to aX$$

$$X \to aY$$

$$Y \to b \mid aXB$$

$$B \to b$$

5.

$$S \to AB \mid aA \mid bB \mid a \mid b \mid cC \mid c \mid d$$

$$A \to aA \mid bB \mid a \mid b$$

$$B \to bB \mid b$$

$$C \to cC \mid c \mid d$$

6.

a.

$$\begin{aligned} S &\to X_aD_1 \mid X_aD_2 \\ A &\to X_bS \mid X_aX_a \\ B &\to X_aB \mid b \\ C &\to BA \mid X_bX_b \\ D_1 &\to BA \\ D_2 &\to X_bC \\ X_a &\to a \\ X_b &\to b \end{aligned}$$

b.

$$\begin{aligned} S &\to aBA \mid aX_bC \\ A &\to bS \mid aX_a \\ B &\to aB \mid b \\ C &\to aBA \mid bA \mid bX_b \\ X_a &\to a \\ X_b &\to b \end{aligned}$$